

Chase Horne

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PROFESSIONAL SUMMARY

Computer Science graduate (B.S., West Virginia University, May 2026) with experience in systems programming, backend development, and embedded software: a multitasking kernel and heap allocator in C, a Flask REST API over a normalized MySQL schema, and an Arduino tachometer validated against a stroboscope. Seeking a full-time software engineering role.

TECHNICAL SKILLS

Languages: C, Python, Java, JavaScript, TypeScript, SQL

Backend & Data: Flask, REST APIs, MySQL, schema design, parameterized queries, data structures & algorithms

Systems & Embedded: Process scheduling, memory allocation, serial I/O, Arduino / embedded C++, QEMU

Web & Tools: React, HTML/CSS, Git, GitHub, Linux, VS Code, Xcode, unit testing

PROJECTS

Multitasking Operating System — C, x86, QEMU | team of 4 Jan – May 2025

- Built process control blocks and five priority-ordered scheduling queues, making dispatch of the next process constant time.
- Implemented a first-fit heap allocator with Memory Control Blocks, splitting blocks on allocation and coalescing them on release.
- Wrote the command handler and serial driver; debugged bare-metal kernel state over the serial port.
- Fixed an allocator bug where pointer arithmetic advanced in struct-size units, not bytes, preventing reuse of freed memory.

Mad Bracket — Sports Bracket Web Application — Python, Flask, MySQL | team of 4 Spring 2026

- Designed a normalized five-table MySQL schema (conferences, teams, stats, coaches, brackets) with foreign-key constraints.
- Built the Flask REST API: a three-table JOIN returning a team record and conference in one request, plus CRUD for saved brackets.
- Used parameterized queries throughout; loaded database credentials from environment variables rather than source.

Optical RPM Measurement System — Arduino, C++ March 2026

- Built an optical tachometer with a TCRT5000 reflective sensor to measure motor speed in real time.
- Implemented state-based edge detection so each marker pass counts once, not on every sample above threshold.
- Calibrated the detection threshold and sampling window against a stroboscope, validating readings to 5,000 RPM.

Tackle App — Fishing Conditions Web App — JavaScript, React, REST APIs | team of 6 Fall 2024

- Built the weather conditions component, integrating the OpenWeather API for day-by-day conditions by location.
- Designed front-end data-visualization components, improving readability of live marine data

EDUCATION

West Virginia University — Morgantown, WV May 2026

B.S. in Computer Science, Minor in Communications | GPA: 3.28

Relevant Coursework: Operating Systems, Data Structures & Algorithms, Database Design, Artificial Intelligence

EXPERIENCE

Manager — Chick-fil-A — Pittsburgh, PA Feb 2021 – Present

- Train and manage 20–30 team members across 10 stations in a high-volume service environment.
- Coordinate scheduling, onboarding, and performance feedback; resolve operational issues during peak service.

Teaching Assistant — West Virginia University — Morgantown, WV Aug – Dec 2025

- Worked one-on-one with students debugging code and reinforcing core computer science concepts.

LEADERSHIP & ACTIVITIES

Membership Chair, National Society of Black Engineers (NSBE) Aug 2023 – Present

Volunteer, Chick-fil-A Kids Marathon May 2023